Fleet Assignment: Complete Optimization Model

Author: Nicolás Cortines, Chun-An Ko, Chongyang He, Buwei Xiao, Erica Wang

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1 Problem overview

Problem. Assign one eligible aircraft type to each flight in a fixed one-day schedule, so that capacity matches demand at minimum cost.

- **Objective:** minimize total cost, defined as the sum of:
  - **Operating cost (OpEx):** cost per airborne hour of the aircraft type × flight airborne time;
  - **Spill cost:** opportunity cost of passenger demand that exceeds the seating capacity of the assigned aircraft (spilled passengers × average fare).
- **Trade-off:** larger aircraft reduce spill but increase OpEx; smaller aircraft reduce OpEx but leave demand unserved.

Assumptions

1. **Representative-day Schedule.** The October 16, 2024 schdule is treated as a daily operating pattern. Aircraft locations at the end of the day carry over to the beginning of the next day. The model therefore uses cyclic aircraft flows rather than a separately specified initial aircraft distribution at each airport.
2. **Type-dependent turnaround.** Each aircraft type has a deterministic minimum turnaround time, assumed to be the same at every airport. An aircraft becomes available after completing its flight and the full turnaround time. It may operate another flight departing at the exact time it becomes available.
3. **No repositioning flights.** Other unscheduled repositioning flights are not added in this version. The aircraft can move between airports only by operating a flight in the scheduled flight set.
4. **Deterministic demand and lost spill.** Demand is treated as deterministic. Passengers who cannot be accommodated on the assigned aircraft are treated as spilled, without recapture on other flights. The spill cost is approximated by the estimated average fare of the flight.
5. **Fixed Operations.** The schedule does not change. The model does not include cancellations, delayed propagation, maintenance, or airport and gate capacity constraints.

2 Generalization

2.1 Sets and Parameters

Replacing the example's values with symbols extends the model to all flights. The formulation follows integer-programming fleet assignment [1, 4], building on classic aircraft-allocation models [3], with a flight-level simplification of itinerary-based spill models [2].

Sets.

- A: airports, indexed by a.
- F: domestic flights operated by Hawaiian Airlines on October 16, 2024; f indexes flights.
- $K = \{608, 696, 721, 889\}$: candidate aircraft types, identified by their BTS T-100 codes: 608 (Boeing 717), 696 (Airbus A330), 721 (Airbus A321neo) and 889 (Boeing 787); k indexes aircraft types.
- $K_f \subseteq K$: aircraft types eligible for flight f, i.e. types that operated the same directional route in the October 2024 T-100 data (PHX–HNL and HNL–PHX are different routes).

Flight Parameters.

Symbol	Meaning	Data Source/Calculation
o_f	Origin airport of flight f	BTS On-Time Performance
d_f	Destination airport of flight f	BTS On-Time Performance
τ_f^{dep}	Scheduled departure time of flight f, in hours	BTS On-Time Performance; converted to the common time basis
τ_f^{arr}	Scheduled arrival time of flight f, in hours	BTS On-Time Performance; converted to the common time basis
h_f	Estimated airborne time of flight f, in hours	$\frac{\text{T-100 route-level AIR TIME}}{60\times\text{performed departures}}$
q_f	Estimated demand of flight f	$\frac{\text{T-100 passengers}}{\text{performed departures}} \times \phi_f$,when the load-factor rule call for an adjustment
p_f	Estimated Penalty per spilled passenger on flight f	Average DB1B fare for the flight's directional route
ϕ_f	Demand adjustment factor, with $\phi_f \geq 1$	Team-defined estimation input; document the rule for applying it

Aircraft-type Parameters.

Symbol	Meaning	Data Source/Calculation
v_k	Seating capacity of aircraft type k	$\frac{\text{T-100 Seats}}{\text{performed departures for type } k}$
c_k	Operating cost per airborne hour for type k	Form 41 Schedule P-5.2 $\frac{\text{operating expenses}}{\text{airborne hours for type } k}$
t_k	Minimum turnaround time for k	Operational data or a documented modeling assumption
n_k	Number of available aircraft of type k	Airline fleet inventory or another documented fleet-count source

The listed BTS and fare datasets do not directly provide the turnaround times or fleet counts required by the model. The team must document the sources or assumptions used for t_k and n_k. Any proxy used for an aircraft type without a direct operating-cost estimate must also be documented.

All times are converted to a common time basis and expressed in hours. The time unit for the schedule must be consistent with the units of the turnaround times.

2.2 Time-Space Network Derived from the Fixed Data

For any time u, let $\langle u \rangle_{24}$ denote its time of day in the interval $[0, 24)$:

$$\langle u \rangle_{24} = u - 24 \left\lfloor \frac{u}{24} \right\rfloor .$$

Define the event-time set

$$\mathcal{T} = \text{sort_unique}\Big(\{ \langle \tau_f^{\text{dep}} \rangle_{24} : f \in \mathcal{F} \} \cup \{ \langle \tau_f^{\text{arr}} + t_k \rangle_{24} : f \in \mathcal{F},\ k \in \mathcal{K}_f \} \Big),$$

where

$$\mathcal{T} = \{\theta_1, \ldots, \theta_\mu\}, \qquad \theta_1 < \cdots < \theta_\mu, \qquad \mu = |\mathcal{T}|.$$

Thus, the event-time set contains all scheduled departures and all possible aircraft-ready times. It is shared by all airports.

For airport a, type k, and event time θ_m, define

$$\mathcal{I}_k(a, \theta_m) = \left\{ f \in \mathcal{F} : k \in \mathcal{K}_f,\ d_f = a,\ \langle \tau_f^{\text{arr}} + t_k \rangle_{24} = \theta_m \right\},$$

and

$$\mathcal{O}_k(a, \theta_m) = \left\{ f \in \mathcal{F} : k \in \mathcal{K}_f,\ o_f = a,\ \langle \tau_f^{\text{dep}} \rangle_{24} = \theta_m \right\}.$$

The set $\mathcal{I}_k(a, \theta_m)$ contains flights whose type-k aircraft complete turnaround and become available at airport a at event time θ_m. The set $\mathcal{O}_k(a, \theta_m)$ contains flights departing airport a at that event time.

Let $z \in \mathbb{Z}$ index daily copies of the representative schedule. Define

$$\mathcal{B}_k(\theta_m) = \left\{ (f,z) \;\middle|\; \begin{array}{l} f \in \mathcal{F}, \quad k \in \mathcal{K}_f, \quad z \in \mathbb{Z}, \\ \tau_f^{\text{dep}} + 24z \le \theta_m < \tau_f^{\text{arr}} + t_k + 24z \end{array} \right\}.$$

The set $\mathcal{B}_k(\theta_m)$ contains the flight copies whose type-k aircraft are still flying or completing turnaround at event time θ_m. The interval includes the departure time and excludes the time when turnaround is complete.

These sets are derived from the fixed schedule and aircraft data. They identify where and when aircraft become available, depart, or remain committed to a flight and its turnaround.

2.3 Decision Variables and Model-Derived Spill

For each eligible flight-type pair, define the assignment variable

$$x_{fk} = \begin{cases} 1, & \text{if flight } f \text{ is assigned to aircraft type } k, \\ 0, & \text{otherwise,} \end{cases} \qquad f \in \mathcal{F}, \quad k \in \mathcal{K}_f.$$

For each airport, aircraft type, and event time, define the inventory variable

$$\begin{array}{rl} y_{akm} = \text{number of type-}k \text{ aircraft at airport } a & a \in \mathcal{A}, \quad k \in \mathcal{K}, \\ \text{just after event time } \theta_m, & m = 1, \ldots, \mu. \end{array}$$

For each flight, define u_f as the number of passengers spilled from that flight. Given the assignment decision, the selected seat capacity is

$$\sum_{k \in \mathcal{K}_f} v_k x_{fk}.$$

Therefore, the spill implied by the assignment is

$$u_f = \max \left\{ q_f - \sum_{k \in \mathcal{K}_f} v_k x_{fk},\ 0 \right\}, \qquad f \in \mathcal{F}.$$

This formula describes the intended spill calculation. The MILP represents it using linear inequalities in Section 6.

2.4 Objective Function

The model minimizes total aircraft operating cost plus estimated passenger-spill cost:

$$\min_{x,y,u} \quad \sum_{f \in \mathcal{F}} \sum_{k \in \mathcal{K}_f} c_k h_f x_{fk} + \sum_{f \in \mathcal{F}} p_f u_f.$$

The first term charges the selected aircraft type's cost per airborne hour multiplied by the flight's airborne time. The second term charges the estimated fare for each spilled passenger.

3 Constraints and Their Operational Meaning

3.1 Flight coverage

$$\sum_{k \in \mathcal{K}_f} x_{fk} = 1, \qquad \forall f \in \mathcal{F}.$$

Each scheduled flight must be assigned exactly one eligible aircraft type.

3.2 Linear spill calculation

$$u_f \geq q_f - \sum_{k \in \mathcal{K}_f} v_k x_{fk}, \qquad \forall f \in \mathcal{F},$$
$$u_f \geq 0, \qquad \forall f \in \mathcal{F}.$$

These inequalities impose the positive-part spill calculation from Section 4 in linear form. Since the objective minimizes $p_f u_f$ and $p_f > 0$, the optimal u_f equals the demand shortfall when demand exceeds capacity, and zero otherwise. Thus, the max expression is represented without a nonlinear constraint.

3.3 Cyclic boundary

$$y_{ak,0} = y_{ak,\mu}, \qquad \forall a \in \mathcal{A}, \quad \forall k \in \mathcal{K}.$$

For notation in the flow balance, $y_{ak,0}$ refers to the inventory after the final event of the previous representative day. This boundary condition connects consecutive daily cycles.

3.4 Airport aircraft-flow balance

$$y_{ak,m-1} + \sum_{f \in \mathcal{I}_k(a,\theta_m)} x_{fk} = \sum_{f \in \mathcal{O}_k(a,\theta_m)} x_{fk} + y_{akm},$$
$$\forall a \in \mathcal{A}, \quad \forall k \in \mathcal{K}, \quad m = 1, \ldots, \mu.$$

At each airport and event time, aircraft carried over from the previous ground interval plus aircraft completing turnaround must equal aircraft departing plus aircraft remaining on the ground. Because the inbound set is indexed by airport, aircraft type, and event time, this constraint prevents a flight from using an aircraft that is unavailable at its origin. It also permits an aircraft completing its full turnaround at an event time to depart at that same time.

3.5 Fleet-size limit

$$\sum_{a \in \mathcal{A}} y_{akm} + \sum_{(f,z) \in \mathcal{B}_k(\theta_m)} x_{fk} \leq n_k, \qquad \forall k \in \mathcal{K}, \quad m = 1, \ldots, \mu.$$

At every event time, the number of type-k aircraft waiting at airports plus the number still flying or completing turnaround cannot exceed the available fleet. Aircraft counts do not change between consecutive events, so checking the event times is sufficient.

4 Variable Domains

The variable domains are

$$x_{fk} \in \{0, 1\}, \qquad \forall f \in \mathcal{F}, \quad k \in \mathcal{K}_f,$$
$$y_{akm} \in \mathbb{Z}_{\geq 0}, \qquad \forall a \in \mathcal{A}, \quad \forall k \in \mathcal{K}, \quad m = 1, \ldots, \mu,$$
$$u_f \in \mathbb{R}_{\geq 0}, \qquad \forall f \in \mathcal{F}.$$

The spill variable is continuous because the demand values are route-level estimates and may be fractional. Aircraft inventories are nonnegative integers.

5 Consolidated MILP Formulation

Using the definitions above, the complete model is

$$
\begin{aligned}
\min_{x,y,u} \quad & \sum_{f\in\mathcal{F}}\sum_{k\in\mathcal{K}_f} c_k h_f x_{fk} + \sum_{f\in\mathcal{F}} p_f u_f \\
\text{s.t.} \quad & \sum_{k\in\mathcal{K}_f} x_{fk} = 1, \qquad \forall f\in\mathcal{F}, \\
& u_f \geq q_f - \sum_{k\in\mathcal{K}_f} v_k x_{fk}, \qquad \forall f\in\mathcal{F}, \\
& u_f \geq 0, \qquad \forall f\in\mathcal{F}, \\
& y_{ak,0} = y_{ak,\mu}, \qquad \forall a\in\mathcal{A},\ k\in\mathcal{K}, \\
& y_{ak,m-1} + \sum_{f\in\mathcal{I}_k(a,\theta_m)} x_{fk} = \sum_{f\in\mathcal{O}_k(a,\theta_m)} x_{fk} + y_{akm}, \\
& \qquad \forall a\in\mathcal{A},\ k\in\mathcal{K},\ m=1,\dots,\mu, \\
& \sum_{a\in\mathcal{A}} y_{akm} + \sum_{(f,z)\in\mathcal{B}_k(\theta_m)} x_{fk} \leq n_k, \\
& \qquad \forall k\in\mathcal{K},\ m=1,\dots,\mu, \\
& x_{fk}\in\{0,1\}, \qquad \forall f\in\mathcal{F},\ k\in\mathcal{K}_f, \\
& y_{akm}\in\mathbb{Z}_{\geq 0}, \qquad \forall a\in\mathcal{A},\ k\in\mathcal{K},\ m=1,\dots,\mu.
\end{aligned}
$$

The consolidated formulation uses the previously defined notation; it does not introduce additional sets, parameters, or variables. Because repositioning flights are excluded, infeasibility would mean the schedule cannot be covered under the specified eligibility, timing, turnaround, and fleet inputs without additional aircraft or repositioning.

AI use disclosure

Generative AI assisted with the model formulation, data-source selection, explanations and LaTeX formatting.

References

[1] J. Abara. Applying integer linear programming to the fleet assignment problem. *Interfaces*, 19(4):20–28, 1989.

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[4] H. D. Sherali, E. K. Bish, and X. Zhu. Airline fleet assignment concepts, models, and algorithms. *European Journal of Operational Research*, 172(1):1–30, 2006.